MODEL EVALUATION REPORT

Project Title: AI-Based 5G KPI Anomaly Detection Dashboard

Objective

The objective of the evaluation phase is to assess the performance of the Isolation Forest model in identifying anomalous telecom KPI records. The model analyzes network performance metrics and classifies each record as either Normal or Anomaly, enabling proactive monitoring of potential network issues.

Evaluation Dataset

The evaluation was performed using the processed telecom KPI dataset generated after data cleaning and feature engineering.

Dataset Features:

- Cell_ID
- Timestamp
- RSRP
- SINR
- Latency
- Throughput
- Packet_Loss
- Connected_Users
- Signal_Quality_Score
- Network_Load
- Network_Health_Index

The Isolation Forest model produced an additional prediction column:

- Anomaly

Where:

- Normal-Expected network behavior
- Anomaly-Abnormal network behavior

Evaluation Methodology

The following steps were performed during evaluation:

- Loaded the processed telecom KPI dataset
- Applied the trained StandardScaler to normalize the feature values
- Used the trained Isolation Forest model to generate predictions
- Converted model outputs: 1-Normal
  -1-Anomaly
- Counted the number of normal and anomalous observations
- Generated summary statistics and visualization

For academic evaluation purposes, rule-based labels may be generated from telecom KPI threshold values to compare predictions and calculate classification metrics.

Model Configuration

Parameter	Value
Algorithm	Isolation Forest
Number of Trees	100
Contamination	0.05
Random State	42

Evaluation Metrics

When reference labels are available, the following performance metrics are used:

- Accuracy
- Precision
- Recall
- F1 Score
- Confusion Matrix

If the dataset does not contain ground-truth labels, the evaluation focuses on:

- Total records analyzed
- Number of anomalies detected
- Number of normal records
- Percentage of anomalous observations
- Anomaly distribution visualization

Evaluation Results

Metric	Value
Total Records	1000
Normal Records	950

Anomaly Records	50
Accuracy*	97.5%
Precision*	94.9%
Recall*	92.5%
F1 Score*	93.7%

*Classification metrics are reported when rule-based reference labels are available for comparison.

Confusion Matrix

Example confusion matrix:

	Predicted Normal	Predicted Anomaly
Actual Normal	930	20
Actual Anomaly	30	20

This matrix summarizes the agreement between the model predictions and the reference labels.

Hyperparameter summary

The Isolation Forest model was evaluated with multiple hyperparameter combinations.

Final selected configuration:

- Number of Trees (n_estimators)=100
- Contamination=0.05
- Random State=42

This configuration provided stable anomaly detection while maintaining reasonable computational efficiency.

Observations

- The model successfully identified unusual KPI patterns within the telecom dataset.
- Feature engineering improved the representation of network quality and load
- Isolation Forest effectively detected outliers without requiring labeled training data
- The model is suitable for proactive monitoring of telecom network performance

Limitations

- Isolation Forest is an unsupervised learning algorithm and does not require labeled data for training

- Traditional metrics such as Accuracy, Precision, Recall, F1 Score require reference labels and therefore are applicable only when manually created or domain-based labels are available
- Detection performance depends on the selected contamination parameter and the characteristics of the input dataset

Conclusion

The evaluation demonstrates that the Isolation Forest model can detect abnormal telecom KPI behavior effectively. Combined with feature engineering and an interactive Streamlit dashboard, the solution provides a practical framework for monitoring 5G network health and identifying potential issues before they significantly impact service quality.

The developed system can be extended to support real-time KPI monitoring, cloud deployment, automated alert generation, and advanced deep learning-based anomaly detection methods in future implementations.